TOR AMPLITUDE IS PRESERVED BY DERIVED PULLBACK

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites the lemmas recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Tor amplitude characterization. Cohomology of Sheaves, Lemma cohomology-lemma-tor-amplitude. Flat representative in the range.
- Pullback of flat modules. Modules, Lemma modules-lemma-base-change-flat. f^*E^i is flat.

Source reference. Original lemma 08CI: sources/stacks-chow/tags.txt, source lines 10603–10603. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 0H97: sources/stacks-chow/tags.txt, source lines 21517–21517. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Let E be an object of $D(\mathcal{O}_Y)$. If E has tor amplitude in $[a, b]$, then Lf^*E has tor amplitude in $[a, b]$.*

Proof. Assume E has tor amplitude in $[a, b]$. By Lemma 08CI we can represent E by a complex of $\mathcal{E}^\bullet$ of flat $\mathcal{O}$-modules with $\mathcal{E}^i = 0$ for $i \notin [a, b]$. Then Lf^*E is represented by $f^*\mathcal{E}^\bullet$. By Modules, Lemma 0H97 the modules $f^*\mathcal{E}^i$ are flat. Thus by Lemma 08CI we conclude that Lf^*E has tor amplitude in $[a, b]$. □

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.